

RITESH SAHANI

Computer Engineer | Full-Stack & AI/ML

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ABOUT ME

I'm a recent Computer Engineering graduate from Tribhuvan University, IOE Purwanchal Campus. I have strong foundations in OS, DBMS, DSA, networking, and AI/ML, paired with practical experience building systems using modern tech stacks. A motivated, quick-learning team player seeking internship or entry-level software engineering or IT roles.

EDUCATION

Bachelor in Computer Engineering

TU, IOE Purwanchal Campus, Dharan

2083 BS

Percentage : 72.98 %

Intermediate Level Education (TSLC) - Technical & Vocational Stream

Shree Durga Secondary School, Kalaiya

2077 BS

CGPA: 3.44

EXPERIENCE

C Programming Exam Preparation Tutor | [IOE Purwanchal Campus, Dharan](#) (Organized by BPR)

Duration: 1 week

→ Delivered targeted C programming lectures and intensive problem-solving sessions to help students prepare for upcoming engineering board examinations.

On-the-Job Trainee – Java Application Development | [Bara Computer Institute, Kalaiya](#)

Duration: 1 month

→ Gained practical experience with Java fundamentals, OOP principles, and basic application development.

On-the-Job Trainee – WordPress Site Development | [NIIT Birgunj](#)

Duration: 6 months

→ Developed, launched, and managed responsive websites by configuring custom WordPress themes, plugins, and core layout architectures.

→ Optimized page load speeds and user interfaces across multiple client websites through asset compression and clean CSS styling.

PROJECTS

Nepali Semantic Search and RAG Based Chat System for Legal Document

Improved search accuracy metrics—MRR@10 by 21.47%, NDCG@10 by 18.44%, and MAP@100 by 20.92%—by fine-tuning the intfloat/multilingual-e5-large model using PyTorch, Hugging Face, and public datasets.

Image Colorization Using GAN

Achieved a Peak Signal-to-Noise Ratio (PSNR) of 25 DB by implementing a two-stage training strategy (generator pre-training followed by joint adversarial training) for a cGAN.

Real Time Chat Application

Architected real-time, bi-directional chat system using NodeJs and Socket.io .

TECHNICAL SKILLS AND FAMILIARITY

Languages: C, C++, JavaScript, Python, Java, PHP

AI / ML: PyTorch, Hugging Face Transformers, LangChain, scikit-learn, numpy, pandas, matplotlib

Web & Frameworks: HTML, CSS, ReactJS, NodeJs, Django, Laravel

Tools & DevOps: Git, GitHub, VS Code, Postman, Docker (basic)

Databases: MySQL, MongoDB

ADDITIONAL INFORMATION

Languages: Bhojपुरी | Nepali | English | Hindi

Soft Skills: Teamwork, Communication, Leadership, Problem Solving